

Research highlights

- Proposes bi-level reinforcement learning for joint bus planning and holding.
- Coordinates asynchronous dispatch and stop decisions with goal-conditioned coupling.
- Adapts fleet and headway penalties through online and Lagrangian learning.
- Reduces composite cost by 8.2% over fixed timetables and 4.7% over CMA-ES.
- Produces fleet-budget Pareto fronts from one policy under stochastic demand.